

Khushi Patel

706-980-7352 | ksptl2005@gmail.com | <https://www.linkedin.com/in/khushispat/> | [khuxhip.github.io/](https://github.com/khuxhip)

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. Mechanical Engineering, Minors in Computer Science & Health Technology

May 2027

- **Coursework:** Dynamics of Modeling Systems, Materials, Fluid Dynamics, Electronic Circuits, Analog Signal Processing, Data Structures & Algorithms, Database Systems, Manufacturing Design, Mechanical Design, Solid Mechanics, Statics

WORK EXPERIENCE

Abbott

Multiple Locations

Structural Heart Intern

Jun 2026 – Present

- Support manufacturing operations and process improvement for Abbott's Structural Heart eValve team

ANRA Systems Intern

Sep 2025 – Jun 2026

- Audited 180+ SharePoint pages and resolved document-control and content-integrity issues
- Analyzed review workflows across 9 regulatory-data systems and developed a centralized tracker to automate content reviews
- Converted user requirements into workflows and technical documentation supporting regulatory data and process compliance

Nutrition Engineering Intern

Jun 2025 – Aug 2025

- Performed manufacturing quality checks across ingredient and production stages to verify product compliance and reliability
- Designed an ingredient qualification workflow used by 20+ engineers, reducing processing time by 50%
- Built a Python analytics pipeline automating 2 supplier analyses across Abbott's ingredient portfolio

Speech Perception Laboratory

Champaign, IL

Undergraduate Research Assistant

Jan 2025 – Present

- Researched ML techniques to analyze speech intelligibility in noisy environments and integrated LLMs to assess speech understanding
- Analyzed clinical speech-intelligibility datasets from 100+ participants to compare model performance using perplexity
- Applied RoBERTa-based perplexity analysis at 3 linguistic levels to compare difficulty across QuickSIN test lists

LEADERSHIP

American Society of Mechanical Engineers

Champaign, IL

Director of Product Development

May 2025 – May 2026

- Led 50+ members developing hardware solutions for industry and academic clients, including a Y Combinator-selected startup
- Directing projects including assistive devices, a 3D printing robotic arm, and other technologies
- Organizing and teaching FEA & DFM workshops to 100+ students, supporting technical training

Intelligent Mattress Topper Lead Engineer

Oct. 2024 – Aug. 2025

- Led a six-member team developing a smart mattress topper for continuous pressure monitoring and risk identification by 60%
- Built a real-time Python heatmap from a 16-sensor Arduino array to visualize pressure distribution
- Designed scalable embedded PCBs in KiCad and an air-valve repositioning system

PROJECTS

Heptathlon Car | FEA, Fusion360, GD&T

Jan 2026 – May 2026

- Designed an IR-controlled RC car with a 3-mode, servo-actuated sliding-mesh transmission with a team
- Validated TPU roof under a 3 kg drop, achieving a 17.15 safety factor with peak stress 97.3% below yield strength
- Simplified the transmission from four speeds to three modes, reducing complexity by 25%

Cough-E | FEA, PVA, Fusion360

Aug 2025 – Dec 2025

- Co-designed to design a 300:1 dispenser gear train with a walking-speed of 4.35 m/min
- Performed PVA on a six-bar walking linkage, determining a 35% duty cycle and 4.375 cm stride length
- Reduced the active linkage-leg count by 33%, synchronizing two legs in phase with a support wheel

MeloMood | Python, SQL, JavaScript

May 2025 – Aug 2025

- Developed a full-stack music recommendation platform with RESTful APIs, & mood analytics for 100+ users
- Integrated the Spotify Web API to generate 10-song playlists based on mood logs and listening history

Hydraulic Claw Machine | DFM, CAD, GD&T

Aug 2023 – May 2024

- Collaborated with team members to build a full-scale hydraulic claw machine with 4 syringe-operated controls for horizontal, vertical, and claw actuation

MedSecure | SolidWorks, Arduino, Embedded Systems, C/C++

Aug 2022 – Jul 2023

- Co-developed a 12 × 12 × 12 in. automated medication dispenser integrating motorized delivery, Arduino control, Wi-Fi, pulse sensing, and a 16 × 2 LCD with a team of 4

TECHNICAL SKILLS

Software: SolidWorks, OnShape, Fusion360, KiCAD, MATLAB/Simulink, Arduino, Cura

Technical: Geometric Dimensioning & Tolerancing (GD&T), Finite Element Analysis (FEA), Computer Aided Design (CAD), Design for Manufacturing (DFM), Soldering, Circuit Design, Machine Learning

Programming: C/C++, Python, SQL, HTML/CSS, MongoDB, Flask, Java, JavaScript